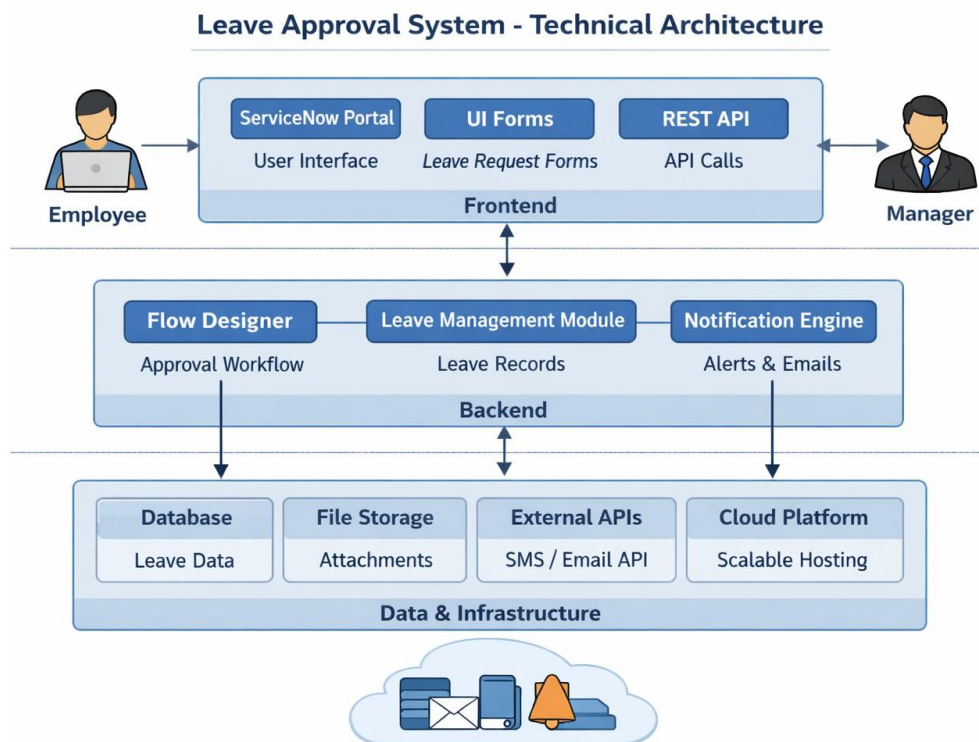


## Requirement Analysis

### Technology Stack (Architecture & Stack)

|               |                       |
|---------------|-----------------------|
| Date          | 30 June 2026          |
| Roll Number   | 23A51A4279            |
| Project Name  | Leave Approval System |
| Maximum Marks | 4 Marks               |

### Technical Architecture (Leave Approval System)



**Table 1: Components**

| S.No | Component           | Description                                                   | Technology                           |
|------|---------------------|---------------------------------------------------------------|--------------------------------------|
| 1    | User Interface      | Employee interacts via Web UI or Mobile App                   | HTML, CSS, JavaScript, Angular/React |
| 2    | Application Logic-1 | Leave request submission & validation (end $\geq$ start date) | JavaScript / ServiceNow Scripting    |
| 3    | Application Logic-2 | Workflow routing for manager approval                         | ServiceNow Flow Designer             |
| 4    | Application Logic-3 | Notification logic for approval/rejection                     | ServiceNow Notification Engine       |

| S.No | Component      | Description                                     | Technology                             |
|------|----------------|-------------------------------------------------|----------------------------------------|
| 5    | Database       | Stores leave requests, approvals, and reports   | MySQL / ServiceNow Tables              |
| 6    | Cloud Database | Cloud storage for scalability                   | ServiceNow Cloud DB / AWS RDS          |
| 7    | File Storage   | Stores attachments (medical certificates, etc.) | ServiceNow Attachments / Cloud Storage |
| 8    | External API-1 | Email/SMS notification integration              | Twilio API / SMTP                      |
| 9    | External API-2 | HR system integration for employee records      | REST API                               |
| 10   | Infrastructure | Deployment on ServiceNow Cloud                  | ServiceNow PaaS / Kubernetes           |

**Table 2: Application Characteristics**

| S.No | Characteristics          | Description                                      | Technology                            |
|------|--------------------------|--------------------------------------------------|---------------------------------------|
| 1    | Open-Source Frameworks   | UI frameworks for responsive design              | Angular, React                        |
| 2    | Security Implementations | Role-based access, encryption, IAM controls      | SHA-256, HTTPS, ServiceNow ACLs       |
| 3    | Scalable Architecture    | Microservices & cloud-based deployment           | ServiceNow Cloud, Kubernetes          |
| 4    | Availability             | High availability with load balancing            | ServiceNow Cloud, AWS Load Balancer   |
| 5    | Performance              | Optimized workflows, caching, fast notifications | ServiceNow Performance Analytics, CDN |